



Entri Elevate

FINAL PROJECT

E-COMMERCE REVENUE PREDICTION ANALYSIS

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ABSTRACT

E-commerce platforms generate large volumes of data related to customer activity, website traffic, marketing expenditure, orders, discounts, customer behavior, and operational performance. Analyzing these factors can provide valuable insights into revenue generation and support better business decision-making.

The project **“E-Commerce Revenue Analysis and Prediction”** focuses on analyzing an e-commerce dataset containing 6,000 records collected over the period from 2022 to 2025. The project involves data understanding, data selection, data cleaning, data transformation, feature engineering, exploratory data analysis, statistical analysis, visualization, and preparation for predictive modeling.

The analysis examines the relationship between revenue and factors such as marketing expenditure, website visits, orders, average order value, conversion rate, returning customer rate, discounts, advertising activity, customer ratings, product availability, shipping delays, competitor pricing, seasonality, sales channels, and regions.

Several statistical techniques and visualizations were used to identify patterns and relationships in the data. The analysis found that Average Order Value has a particularly strong positive relationship with Revenue, while revenue also varies across sales channels, regions, weekdays and weekends. Feature engineering was performed to create additional analytical variables that represent marketing efficiency, customer behavior, operational performance, and time-based characteristics.

The final cleaned dataset provides a strong foundation for developing machine-learning models for revenue prediction. Future work will focus on training and comparing regression models, evaluating their performance using appropriate metrics, and developing a reliable revenue prediction system.

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CHAPTER 1

INTRODUCTION

The rapid growth of e-commerce has transformed the way businesses sell products and interact with customers. E-commerce platforms generate large volumes of data through customer visits, orders, sales channels, marketing activities, discounts, shipping operations, product availability, customer ratings, and regional sales. Analysing this data can help organizations understand their business performance, identify important factors affecting revenue, and make data-driven decisions.

Revenue is one of the most important performance indicators for an e-commerce business. However, revenue does not depend on a single factor. It can be influenced by several variables such as the number of orders, average order value, website visits, conversion rate, marketing expenditure, discounts, customer behaviour, sales channel, region, product availability, and seasonal patterns. Therefore, systematic analysis of e-commerce data is required to understand the relationships between these factors and overall revenue.

The project **“E-Commerce Revenue Analysis and Prediction”** focuses on analysing historical e-commerce data to identify patterns, trends, relationships, and factors associated with revenue generation. The project follows a structured data science approach involving data selection, data preprocessing, data transformation, feature engineering, exploratory data analysis, statistical analysis, visualization, and predictive modeling.

The dataset used in this project contains e-commerce business records covering the period from **2022 to 2025**. The raw dataset consists of **6,000 records and 23 variables**. Various data preprocessing and feature engineering techniques were applied to prepare the dataset for analysis. Additional features related to time, customer behaviour, marketing performance, order activity, discounts, shipping, and revenue categories were derived from the available data.

Exploratory and statistical analyses are performed to identify significant patterns and relationships within the data. Visualizations are used to present important trends in revenue, orders, sales channels, regions, marketing expenditure, and other business factors in an understandable manner. Statistical techniques are also used to determine whether observed differences and relationships are statistically meaningful.

Finally, predictive modeling is included to investigate whether historical business and operational factors can be used to predict e-commerce revenue. The results of the analysis can provide useful insights for improving marketing strategies, understanding high-performing sales channels and regions, optimizing business operations, and supporting revenue-related decision-making.

Overall, this project demonstrates the application of data science techniques to a real-world e-commerce business problem, from raw data preparation to analysis and predictive modeling.

CHAPTER 2

BUSINESS UNDERSTANDING

Business understanding is the first stage of a data science project. It involves understanding the business context, identifying the main business problem, defining the goals of the analysis, and determining how data-driven methods can support better decision-making.

In an e-commerce business, revenue is influenced by multiple factors such as customer activity, number of orders, average order value, website visits, marketing expenditure, discounts, sales channels, geographical regions, product availability, shipping performance, and customer ratings. Understanding the relationship between these factors is important for evaluating business performance and identifying opportunities for improvement.

The main purpose of this project is to analyse historical e-commerce data and understand the factors associated with revenue generation. The analysis focuses on identifying revenue trends, comparing the performance of different sales channels and regions, examining customer and marketing-related behaviour, and identifying significant relationships among business variables.

The project also aims to apply statistical analysis to determine whether differences observed across categories, such as sales channels, regions, and weekdays versus weekends, are statistically significant. Data visualization is used to communicate important patterns and trends clearly.

In addition to descriptive and statistical analysis, the project includes predictive modeling. The objective of the predictive stage is to determine whether relevant historical business variables can be used to estimate future or expected e-commerce revenue. Model evaluation techniques are then used to assess the performance of the developed predictive models.

The business understanding stage therefore establishes the overall direction of the project and connects the technical data science activities with practical business requirements. The major business questions addressed by the project include:

- What are the overall revenue trends in the e-commerce business?
- Which sales channels generate higher revenue?
- Which geographical regions perform better in terms of revenue?
- How are orders, marketing expenditure, website visits, and average order value related to revenue?

- Are there significant differences in revenue across different business categories?
- How do seasonal and time-based factors affect business performance?
- Can historical business data be used to predict revenue?
- How can the findings support better business and marketing decisions?

By addressing these questions, the project aims to convert raw e-commerce data into meaningful business information and actionable insights that can support revenue analysis, performance monitoring, strategic planning, and data-driven decision-making.

CHAPTER 3

PROBLEM STATEMENT

E-commerce businesses generate large amounts of data from customer interactions, orders, marketing activities, sales channels, geographical regions, and other operational processes. Although this data contains valuable information, raw business data is often difficult to interpret directly and may not clearly reveal the factors that influence revenue.

The major problem addressed in this project is the need to analyse historical e-commerce data to understand revenue performance, identify important factors associated with revenue, and develop a predictive approach for estimating e-commerce revenue.

The available dataset contains information related to orders, revenue, customers, website visits, marketing expenditure, discounts, product availability, shipping performance, customer ratings, sales channels, and geographical regions. Without systematic data preprocessing, transformation, analysis, and visualization, it can be difficult to identify meaningful patterns and relationships within these variables.

Therefore, this project addresses the following key problems:

1. **Data Quality Problem:**

Raw e-commerce data needs to be examined and prepared to ensure that it is suitable for analysis and modeling.

2. **Revenue Analysis Problem:**

The business needs to understand the distribution and overall trends of revenue across different periods and business categories.

3. **Factor Identification Problem:**

It is necessary to identify the business variables that have meaningful relationships with revenue, such as orders, average order value, marketing expenditure, visits, and conversion-related measures.

4. **Performance Comparison Problem:**

The performance of different sales channels and geographical regions needs to be compared to identify high-performing and lower-performing segments.

5. **Statistical Validation Problem:**

Observed differences and relationships need to be statistically examined to determine whether they are meaningful rather than occurring only due to random variation.

6. Prediction Problem:

A predictive modeling approach is required to investigate whether relevant historical business information can be used to estimate e-commerce revenue.

7. Decision-Making Problem:

The analysed results need to be converted into meaningful business insights that can support marketing, sales, operational, and revenue-related decisions.

Hence, the project aims to transform the available e-commerce data into useful information through data preprocessing, feature engineering, exploratory analysis, statistical analysis, visualization, and predictive modeling. The overall goal is to improve the understanding of revenue behaviour and demonstrate how data science can be applied to support data-driven e-commerce decision-making.

CHAPTER 4

OBJECTIVES

The main objective of this project is to analyse e-commerce revenue data using Python-based data analysis techniques and identify meaningful patterns, trends, relationships, and business insights. The specific objectives of the project are:

1. **To understand the e-commerce dataset** by examining its structure, variables, data types, and overall business information.
2. **To clean and preprocess the raw data** by identifying and handling data quality issues such as missing values, duplicate records, incorrect data types, and inconsistent data formats.
3. **To transform the data** into a suitable format for analysis by creating useful date-based and categorical variables.
4. **To perform feature engineering** by deriving meaningful variables from the existing dataset to support deeper analysis of revenue, orders, marketing performance, customer activity, and business operations.
5. **To perform exploratory data analysis (EDA)** using Python to understand the distribution, trends, patterns, and relationships present in the e-commerce data.
6. **To analyse revenue performance** across different sales channels, regions, time periods, and other relevant business factors.
7. **To examine relationships between important variables** such as revenue, orders, average order value, marketing expenditure, website visits, discounts, and conversion-related measures.
8. **To apply statistical analysis techniques** to identify significant relationships and differences among important business variables and categories.
9. **To create meaningful data visualizations** such as bar charts, line charts, scatter plots, and correlation heatmaps to communicate the findings clearly.
10. **To identify important business insights** from the analysis that can help understand revenue performance and support better business decision-making.
11. **To select relevant variables for revenue analysis and prediction** while avoiding inappropriate or directly revenue-derived variables that could lead to misleading results.
12. **To develop and evaluate a predictive model for revenue**, where applicable, using relevant business variables and appropriate evaluation measures.
13. **To present the complete analysis in a structured manner** so that the findings can be easily understood and used for further business analysis.

CHAPTER 5

DATASET DESCRIPTION

The dataset used for this project contains historical e-commerce business information used to analyse revenue performance, customer activity, marketing performance, sales channels, and operational factors. The dataset provides a combination of numerical, categorical, and date-related variables, allowing different types of data analysis to be performed.

The raw dataset contains **6,000 records and 23 variables**. The data covers a period from **1 January 2022 to 31 December 2025**, providing approximately four years of historical e-commerce information.

The main target variable considered in the project is **Revenue**, which represents the revenue generated from the corresponding business activity. Other variables provide information that can be used to understand the factors associated with revenue.

5.1 Dataset Characteristics

Characteristic	Description
Dataset Type	E-commerce business dataset
Number of Records	6,000
Number of Original Variables	23
Time Period	2022–2025
Start Date	1 January 2022
End Date	31 December 2025
Main Analysis Variable	Revenue
Data Types	Numerical, categorical, and date-related
Data Usage	Revenue analysis, feature engineering, statistical analysis, visualization, and prediction

5.2 Major Variables in the Dataset

The dataset contains variables representing different aspects of e-commerce business performance. The major categories of variables are described below.

1. Revenue-related variables:

Revenue is the primary variable used to measure the financial performance of the business. Other order and value-related variables help in understanding revenue generation.

2. Order and customer activity:

Variables related to orders, customers, website visits, and customer activity provide information about purchasing behaviour and business performance.

3. Marketing-related variables:

Marketing expenditure and advertising-related variables are used to examine the relationship between marketing activities and revenue generation.

4. Sales channel variables:

The dataset contains information about different sales channels through which business transactions take place. This allows revenue performance to be compared across channels.

5. Regional variables:

Regional information allows the analysis of revenue and business performance across different geographical regions.

6. Product and operational variables:

Variables related to product availability, discounts, shipping performance, and customer ratings provide additional information about operational and customer-related factors.

7. Date-related information:

The dataset contains date information that allows the creation of additional time-based variables such as year, month, quarter, day of the week, week of the year, and weekend indicators.

5.3 Dataset Structure

The raw dataset was initially examined to understand its structure, data types, completeness, and consistency. The dataset contains **no missing values and no duplicate records**, making it suitable for further preprocessing and analysis. However, data types, date-related fields, and the usefulness of individual variables were examined before performing the subsequent analysis.

During the project, additional variables were created through data transformation and feature engineering. These derived variables were used to support deeper analysis of customer activity, marketing performance, order behaviour, time-based patterns, and revenue characteristics.

The cleaned and transformed dataset therefore contains more variables than the original dataset because several meaningful analytical features were derived from the original 23 variables.

5.4 Purpose of the Dataset

The dataset is used to:

- Analyse historical e-commerce revenue performance.
- Identify trends across different time periods.
- Compare sales performance across channels and regions.
- Examine the relationship between revenue and business variables.
- Analyse marketing and customer activity.
- Create meaningful analytical features.
- Perform statistical analysis.
- Develop informative visualizations.
- Identify important business insights.
- Support revenue prediction using relevant variables.

Thus, the dataset provides the foundation for the complete analysis carried out in this project, from initial data preparation to visualization, statistical analysis, business interpretation, and predictive modeling.

CHAPTER 6

DATA SELECTION

Data selection is the process of identifying the relevant data required to address the objectives of the project. In this project, the available e-commerce dataset was examined to determine which variables were useful for analysing revenue performance and understanding the factors associated with it.

The raw dataset contains **6,000 records and 23 variables** covering the period from **2022 to 2025**. All available variables were initially examined based on their data type, business meaning, completeness, and relevance to the project objectives.

The selection process focused mainly on variables related to the following areas:

- **Revenue and order information** – used to analyse sales and revenue performance.
- **Customer activity** – used to understand customer visits, orders, and purchasing behaviour.
- **Marketing information** – used to examine the relationship between marketing expenditure and revenue.
- **Sales channels** – used to compare revenue performance across different channels.
- **Regional information** – used to analyse geographical differences in business performance.
- **Discount-related information** – used to study the effect of discounts on orders and revenue.
- **Product availability** – used to examine the possible relationship between product availability and business performance.
- **Shipping performance** – used to analyse operational factors associated with revenue.
- **Customer ratings** – used to examine customer-related performance indicators.
- **Date information** – used to analyse revenue and order trends over time.

6.1 Selection Criteria

Variables were selected based on the following criteria:

1. **Relevance to the project objective:** The variable should contribute to understanding revenue or related business performance.
2. **Business significance:** The variable should have a meaningful interpretation in the context of an e-commerce business.
3. **Data availability:** Variables with usable and sufficiently complete data were considered for analysis.

4. **Data type suitability:** Numerical, categorical, and date variables were considered according to the type of analysis required.
5. **Analytical usefulness:** Variables that could support comparisons, correlation analysis, visualization, or feature engineering were retained.

6.2 Target Variable

Revenue was identified as the main variable of interest because the primary purpose of the project is to analyse e-commerce revenue and investigate the factors associated with it.

Revenue was therefore used extensively during exploratory data analysis, statistical analysis, and visualization. For the predictive modeling stage, appropriate independent variables are selected separately to avoid using variables that are directly calculated from Revenue.

6.3 Handling of Derived Variables

During the analysis, several additional variables were created from the original dataset through feature engineering. These variables were not treated as original raw variables. They were evaluated based on their analytical purpose and relevance.

Some derived variables that directly use Revenue in their calculation, such as **Revenue per Order** or other revenue-derived measures, are useful for descriptive analysis but should not be used as independent predictors of Revenue. Excluding such variables from predictive modeling helps avoid **target leakage** and provides a more meaningful assessment of the relationship between the original business factors and revenue.

6.4 Final Data Selection

After examining the available variables, the relevant variables were retained for subsequent stages of the project. The selected data was then passed through data preprocessing and cleaning, followed by transformation, feature engineering, exploratory data analysis, statistical analysis, and visualization.

Therefore, the data selection stage ensures that the analysis is based on variables that are relevant to the project's objectives and have clear business meaning. It also establishes a suitable foundation for the later stages of revenue analysis and prediction.

CHAPTER 7

DATA PRE-PROCESSING AND CLEANING

Data pre-processing and cleaning is an important stage of data analysis that involves examining the raw dataset, identifying potential data quality issues, and preparing the data for further analysis. Proper data cleaning helps ensure that the results obtained from statistical analysis, visualizations, and other analytical techniques are reliable and meaningful.

In this project, the raw e-commerce dataset was examined and processed using Python. The Pandas and NumPy libraries were used to inspect the dataset, examine data types, check data completeness, and perform the required data preparation operations.

The original dataset contained **6,000 records and 23 variables** covering the period from 2022 to 2025. An initial inspection was conducted to understand the dataset's structure and identify any issues that could affect subsequent analysis.

7.1 Initial Data Inspection

The dataset was initially examined to determine the number of records and columns, identify the data types of individual variables, and understand the overall structure of the data. This step helped establish the condition of the raw dataset before further processing.

Python functions such as `head()`, `shape`, `info()`, and `describe()` were used to inspect sample records, dataset dimensions, column information, and descriptive statistics.

7.2 Handling Missing Values

Missing values were checked across all variables to determine whether any records contained incomplete information. The inspection showed that the raw dataset contained **no missing values**.

Therefore, no missing-value imputation or removal was required. The records were retained for subsequent analysis.

7.3 Duplicate Record Identification

The dataset was examined for duplicate records to identify any repeated entries that could affect analytical results. The inspection found **no duplicate records**.

Consequently, no records needed to be removed because of duplication.

7.4 Data Type Inspection

The data types of the variables were examined to ensure that they were suitable for the intended analysis. Numerical variables were reviewed for quantitative analysis, categorical variables were considered for grouping and comparison, and date-related information was examined for time-based analysis.

Correct data types are important because they determine which calculations, grouping operations, statistical methods, and visualizations can be performed appropriately.

7.5 Date Data Preparation

Date-related information was examined to ensure that it could support time-based analysis. The dataset covered the period from **1 January 2022 to 31 December 2025**.

Date information was prepared for subsequent analysis involving years, months, quarters, days of the week, and other time-based characteristics. These variables support the identification of revenue trends and comparisons across different periods.

7.6 Numerical Data Inspection

Numerical variables were examined using descriptive statistics to understand their distributions, ranges, averages, and variability. This helped identify the general characteristics of business variables such as revenue, orders, marketing expenditure, and customer activity.

The examination of numerical variables also provided a foundation for identifying unusual values and understanding the distribution of revenue before conducting more detailed analysis.

7.7 Preparation of the Cleaned Dataset

After the initial inspection and required data preparation, the processed dataset was saved for use in subsequent stages. The cleaned dataset retained the original business records and included additional variables created during later transformation and feature engineering activities.

The resulting dataset contained **6,000 records and 56 variables**, compared with the original 23 variables. The increase in the number of variables resulted from the creation of additional analytical features rather than an increase in the number of records.

7.8 Outcome of Data Pre-processing and Cleaning

The data pre-processing stage established that the raw dataset contained no missing values or duplicate records. The dataset was examined for structural consistency and prepared for further analysis.

The prepared dataset was then used for data transformation, feature engineering, exploratory data analysis, statistical analysis, and visualization. These steps provided a consistent foundation for analysing e-commerce revenue and identifying meaningful business patterns.

CHAPTER 8

DATA TRANSFORMATION

Data transformation is the process of converting and organizing the cleaned data into a suitable format for analysis. After the data pre-processing and cleaning stage, the variables were transformed using Python to make the dataset more useful for identifying time-based patterns, comparing business performance, and performing further analysis.

The transformation process mainly focused on date-related variables, categorical information, and numerical variables. These transformations helped convert the available raw information into meaningful analytical variables without changing the original business records.

8.1 Date Transformation

The date information in the dataset was converted into an appropriate date format to enable time-based analysis. Once the date variable was prepared, different components were extracted from it.

The following date-related variables were created:

- Year – identifies the year of each record.
- Month – represents the numerical month.
- Month Name – represents the month in a readable format.
- Day – represents the day of the month.
- Day of Week – identifies the day on which the record occurred.
- Day of Week Number – provides a numerical representation of the day of the week.
- Quarter – identifies the quarter of the year.
- Week of Year – identifies the week within the year.
- Year-Month – combines year and month for monthly trend analysis.
- Is Weekend – identifies whether the record occurred on a weekend.
- Is Month Start – identifies records occurring at the beginning of a month.
- Is Month End – identifies records occurring at the end of a month.

These transformations allowed revenue and order activity to be analysed across different time periods.

8.2 Categorical Data Transformation

Categorical variables such as sales channel, region, and other business categories were examined and standardized where required. These variables were retained in categorical form because they are useful for grouping, comparison, and visualization.

For example, sales channels can be grouped to compare their average revenue, while regional information can be used to identify differences in business performance across geographical areas.

8.3 Numerical Data Transformation

Numerical variables were examined to ensure that they were stored in appropriate numerical formats and could be used for calculations and statistical analysis. Variables such as revenue, orders, marketing expenditure, visits, discounts, and ratings were prepared for numerical analysis.

Descriptive statistics were used to understand the ranges and distributions of the numerical variables before further analysis.

8.4 Time-Based Aggregation

The transformed date variables were also used to aggregate business information across different periods. Monthly and other time-based summaries were created to analyse changes in revenue and order activity over time.

For example, the Year-Month variable was used to calculate monthly revenue and total orders, allowing trends and changes in business activity to be examined through line charts.

8.5 Purpose of Data Transformation

The main purposes of the transformation process were:

- To make date information suitable for time-based analysis.
- To create meaningful time-related variables.
- To organize categorical variables for comparison.
- To prepare numerical variables for statistical calculations.
- To support grouping and aggregation.
- To make the dataset suitable for visualization.
- To provide a foundation for feature engineering and further analysis.

8.6 Result of Data Transformation

After transformation, the dataset contained additional time-based variables that made it possible to examine revenue and order behaviour from different perspectives. These transformed variables were subsequently used during feature engineering, exploratory data analysis, statistical analysis, and visualization.

Therefore, data transformation converted the cleaned dataset into a more analysis-ready structure while preserving the original business information.

CHAPTER 9

FEATURE ENGINEERING

Feature engineering is the process of creating new and meaningful variables from existing data to support deeper analysis. In this project, feature engineering was performed using Python to derive additional business-related variables from the original e-commerce dataset.

The purpose of feature engineering was not only to increase the number of variables, but also to create measures that provide a better understanding of customer activity, order behaviour, marketing performance, operational performance, and revenue characteristics.

9.1 Order and Customer Activity Features

Several features were created to understand customer and order behaviour. These include measures such as:

- Revenue Per Order – measures the average revenue associated with each order.
- Visitor Order Rate – represents the relationship between orders and website visits.
- Unique Visitor Ratio – compares unique visitors with total visitors.
- Estimated Returning Orders – provides an estimate of orders associated with returning customers.
- Estimated New Customer Orders – provides an estimate of orders associated with new customers.
- Orders Per Ad Click – measures order activity relative to advertising clicks.

These features help provide additional information about customer activity and purchasing behaviour.

9.2 Marketing Performance Features

Marketing-related features were created to examine the efficiency of marketing expenditure. These include:

- Revenue Per Marketing Dollar – measures revenue generated in relation to marketing expenditure.
- Marketing Cost Per Order – measures marketing expenditure associated with each order.
- Marketing Cost Per Visitor – measures marketing expenditure relative to website visitors.
- Marketing Spend Category – groups marketing expenditure into meaningful categories.

These features support the analysis of marketing performance and its relationship with revenue.

9.3 Revenue and Order Value Features

Additional features were created to examine order value and revenue-related characteristics. These include:

- Discount Adjusted Order Value – considers the effect of discounts on order value.
- Potential Order Value – represents an estimated order value before applicable adjustments.
- Revenue Gap – measures the difference between relevant revenue-related values.
- Revenue Category – groups observations into revenue-based categories for descriptive analysis.

Revenue-derived features were primarily used for descriptive and exploratory analysis. Features calculated directly from Revenue were not considered suitable independent variables for predicting Revenue because their use could introduce target leakage.

9.4 Operational Features

Features related to operational performance were also derived to support business analysis. These include:

- Availability Risk – represents the potential effect of product availability on business performance.
- Shipping Performance – provides a measure for analysing shipping-related performance.
- Rating Category – groups customer ratings into meaningful categories.
- Discount Category – groups discount values into categories.

These features help in examining the relationship between operational factors, customer-related indicators, and revenue performance.

9.5 Time-Based Features

The date transformations performed in the previous stage were also used as analytical features. These include:

- Year
- Month
- Month Name
- Quarter
- Day of Week
- Week of Year

- Year-Month
- Is Weekend
- Is Month Start
- Is Month End

These variables allow the analysis to identify seasonal patterns, monthly trends, weekday and weekend differences, and other time-based business behaviour.

9.6 Feature Selection Considerations

Not every engineered feature is suitable for every type of analysis. Features were therefore considered according to their purpose and relationship with the target variable.

For descriptive analysis, revenue-derived features can provide useful information about revenue characteristics. However, when performing predictive modeling, variables that are directly calculated using Revenue should be excluded from the predictor set to avoid target leakage.

The final selection of variables for predictive modeling is therefore based on business relevance and whether the variable would realistically be available independently of the revenue being predicted.

9.7 Outcome of Feature Engineering

Feature engineering increased the analytical information available in the dataset and provided additional measures for examining customer activity, marketing efficiency, order behaviour, operational performance, and time-based patterns.

The engineered features were subsequently used selectively in exploratory data analysis, statistical analysis, visualization, and predictive modeling, depending on their analytical purpose.

Overall, feature engineering helped transform the available e-commerce variables into meaningful measures that support a more detailed analysis of revenue performance.

CHAPTER 10

EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is the process of examining and summarizing a dataset to understand its main characteristics, patterns, distributions, trends, and relationships between variables. In this project, EDA was performed using Python to understand the behaviour of the e-commerce business data before conducting statistical analysis and predictive modeling.

The main purpose of EDA was to explore revenue performance and identify important patterns related to orders, marketing activities, customer visits, sales channels, regions, discounts, and time-based factors. Both numerical summaries and graphical analysis were used during this stage.

10.1 Analysis of Revenue

Revenue was considered the primary variable of interest in the project. Descriptive analysis was performed to understand its central tendency, variation, and distribution.

The analysis showed that the average revenue was approximately 6,879.64, while the median revenue was approximately 5,763.25. The maximum recorded revenue was approximately 50,542.53.

The difference between the mean and median indicates that the revenue distribution is positively skewed, with some observations having considerably higher revenue than the typical observation.

This distribution was considered during subsequent analysis because extreme revenue values can influence statistical measures and relationships between variables.

10.2 Analysis of Orders

Order-related variables were examined to understand the relationship between customer purchasing activity and revenue. The number of orders provides an important indication of business activity and was compared with revenue and other relevant variables.

Order information was also aggregated across different time periods to identify changes in purchasing activity over time.

10.3 Analysis of Sales Channels

The dataset contains information about different sales channels. Revenue was compared across these channels to identify differences in business performance.

The analysis indicated that the Mobile App channel had the highest average revenue among the sales channels considered. This observation provides an indication that the mobile application was an important revenue-generating channel during the period covered by the dataset.

Sales channel performance was further examined using statistical analysis to determine whether the observed differences were statistically significant.

10.4 Analysis of Regional Performance

Regional information was analysed to understand geographical differences in revenue performance.

The analysis showed that the West region had the highest average revenue among the regions considered. This provides a useful business observation for understanding regional revenue performance.

Regional differences were also examined using statistical methods in the subsequent statistical analysis stage.

10.5 Analysis of Marketing Activity

Marketing expenditure and related variables were explored to understand their relationship with revenue.

The analysis indicated a positive relationship between marketing expenditure and revenue, although the strength and interpretation of this relationship were examined more carefully using statistical methods.

Marketing-related features such as marketing cost per order and revenue per marketing dollar were also considered for descriptive analysis.

10.6 Analysis of Average Order Value

Average Order Value (AOV) was examined as an important business performance measure.

The analysis showed a strong positive correlation between Average Order Value and Revenue, with a correlation coefficient of approximately 0.91. This indicates that observations with higher average order values generally tend to have higher revenue.

AOV was therefore considered an important variable when interpreting revenue behaviour.

10.7 Time-Based Analysis

Time-based variables created during the data transformation stage were used to examine revenue and order trends.

Revenue and orders were analysed by year, month, day of the week, quarter, and other time-based variables. Monthly aggregation was particularly useful for identifying changes in business activity over time.

Weekend and weekday performance were also compared. The analysis indicated that weekend revenue was slightly higher than weekday revenue.

10.8 Analysis of Customer and Website Activity

Website visits, customer activity, and order-related variables were explored to understand their relationship with revenue.

Measures such as visitor order rate, unique visitor ratio, and other customer activity features were created during feature engineering. These measures were used to provide additional context when examining customer activity and revenue performance.

10.9 Analysis of Discounts and Operational Factors

Discounts, product availability, shipping performance, and customer ratings were also examined as potential factors associated with business performance.

Categorical features such as discount category and rating category were created to make comparisons easier during exploratory analysis.

These variables were considered as supporting business indicators rather than being interpreted as direct causes of revenue.

10.10 Correlation Analysis

Correlation analysis was used during EDA to identify relationships between numerical variables.

Among the important relationships identified, Average Order Value showed a strong positive correlation with Revenue (approximately 0.91). Other business variables were also examined to understand their direction and strength of association with revenue.

Correlation analysis was used as an exploratory technique and was not interpreted as proof of causation.

10.11 Summary of Exploratory Findings

The major findings from the exploratory analysis include:

- Revenue has a right-skewed distribution.
- The mean revenue is approximately 6,879.64, while the median is approximately 5,763.25.
- The maximum recorded revenue is approximately 50,542.53.
- Average Order Value has a strong positive relationship with Revenue, with a correlation of approximately 0.91.
- The Mobile App has the highest average revenue among the analysed sales channels.
- The West region has the highest average revenue among the analysed regions.
- Weekend revenue is slightly higher than weekday revenue.
- Time-based analysis provides useful information about monthly and seasonal business patterns.
- Marketing, customer activity, discounts, and operational variables provide additional information for understanding revenue behaviour.

The findings obtained from EDA were used as a foundation for the subsequent Statistical Analysis and Data Visualization stages of the project.

CHAPTER 11

STATISTICAL ANALYSIS

Statistical analysis was performed to examine relationships and differences between important e-commerce variables and to support the findings obtained from exploratory data analysis.

Python statistical libraries were used to perform the analysis. The main techniques included correlation analysis, hypothesis testing, and group comparison tests.

The analysis included:

- Correlation Analysis: Used to measure the strength and direction of relationships between numerical variables and Revenue.
- Mann–Whitney U Test: Used to compare Revenue between weekday and weekend observations.
- Kruskal–Wallis Test: Used to examine whether Revenue differed significantly across sales channels and regions.
- Chi-Square Test: Used to examine the association between Sales Channel and Revenue Category.
- Confidence Interval Analysis: Used to estimate the range within which the population mean Revenue is expected to fall.

The statistical analysis showed a strong positive relationship between Average Order Value and Revenue, with a correlation of approximately 0.91. The weekday and weekend comparison also indicated a statistically significant difference in Revenue at the 5% significance level.

Statistical testing further indicated significant differences in Revenue across sales channels and regions. These results provide statistical support for some of the patterns observed during exploratory analysis.

Overall, statistical analysis helped determine whether the observed patterns in the dataset were statistically meaningful rather than relying only on descriptive observations and visualizations.

CHAPTER 12

DATA VISUALIZATION

Data visualization was used to present the results of the e-commerce data analysis in a clear and understandable form. Python visualization libraries, mainly Matplotlib and Seaborn, were used to create charts and graphs.

The visualizations were selected based on the analytical objectives of the project. They were used to understand revenue distribution, compare business categories, identify time-based trends, examine relationships between numerical variables, and communicate important findings.

The major visualizations created in the project include:

1. Revenue Distribution – used to understand the distribution and spread of Revenue.
2. Revenue by Sales Channel – used to compare average revenue across different sales channels.
3. Revenue by Region – used to compare revenue performance across regions.
4. Monthly Revenue Trend – used to identify changes in revenue over time.
5. Monthly Orders Trend – used to analyse changes in order activity over time.
6. Marketing Spend vs Revenue – used to examine the relationship between marketing expenditure and revenue.
7. Average Order Value vs Revenue – used to visualize the strong relationship between AOV and Revenue.
8. Conversion Rate vs Revenue – used to examine the relationship between customer conversion activity and revenue.
9. Revenue Category by Sales Channel – used for descriptive comparison of revenue categories across sales channels.
10. Marketing Spend vs Revenue per Marketing Dollar – used to examine marketing efficiency.
11. Correlation Heatmap – used to display relationships among important numerical variables.

Each visualization was accompanied by an interpretation to explain the pattern observed and its relevance to the business analysis.

The visualizations helped identify important findings such as the right-skewed distribution of Revenue, differences in revenue across sales channels and regions, monthly changes in business activity, and the strong positive relationship between Average Order Value and Revenue.

Overall, data visualization provided an effective way to convert numerical analysis into easily interpretable graphical information and supported the communication of the project's key findings.

CHAPTER 13

BUSINESS INSIGHTS

The analysis of the e-commerce dataset provided several important business insights related to revenue performance, customer activity, sales channels, regions, marketing, and time-based patterns.

13.1 Revenue Performance

The average revenue was approximately 6,879.64, while the median revenue was approximately 5,763.25. The maximum recorded revenue was approximately 50,542.53. The higher mean compared with the median indicates that revenue is right-skewed, with some high-revenue transactions contributing significantly to overall performance.

13.2 Sales Channel Performance

Among the different sales channels, the Mobile App recorded the highest average revenue. This indicates that the mobile channel is an important contributor to revenue and may represent an opportunity for continued investment in mobile customer experience and engagement.

13.3 Regional Performance

The West region recorded the highest average revenue among the regions analyzed. This suggests stronger revenue performance in this region and provides a basis for comparing customer behavior, marketing activity, and sales performance across other regions.

13.4 Average Order Value and Revenue

Average Order Value (AOV) showed a strong positive relationship with Revenue, with a correlation of approximately 0.91. This indicates that transactions with higher order values are strongly associated with higher revenue. Therefore, increasing customer order value can be an important area for revenue improvement.

13.5 Marketing and Revenue

Marketing expenditure showed a positive relationship with revenue. Higher marketing activity was generally associated with higher revenue levels. However, this relationship represents an association and should not be interpreted as direct proof that increased marketing spending alone causes higher revenue.

13.6 Weekday and Weekend Performance

Weekend revenue was slightly higher than weekday revenue. The Mann–Whitney U test indicated that the difference was statistically significant at the 5% significance level. This suggests that customer purchasing behavior differs between weekdays and weekends and that businesses can consider time-based strategies when planning promotions and marketing activities.

13.7 Sales Channel and Regional Differences

Statistical analysis indicated significant differences in revenue across sales channels and regions. This confirms that revenue performance is not uniform across different business segments and that channel- and region-specific analysis can support better decision-making.

13.8 Seasonal and Time-Based Patterns

The time-based analysis showed variations in revenue and order activity across different months, quarters, and periods. The Seasonality Index also showed a positive association with revenue, indicating that time-related factors can contribute to differences in revenue performance.

13.9 Business Implications

Based on the overall analysis, the following actions can be considered:

- Strengthen the Mobile App channel because of its higher average revenue performance.
- Study successful practices in the West region and evaluate whether they can be applied to other regions.
- Focus on strategies that can improve Average Order Value, such as relevant product recommendations and suitable offers.
- Evaluate marketing expenditure together with revenue and marketing efficiency rather than considering spending alone.
- Consider weekend and seasonal patterns when planning campaigns and promotions.

- Monitor revenue performance by channel, region, and time period regularly.
- Use statistical evidence along with visual analysis to support business decisions.

Overall, the analysis demonstrates that revenue performance is influenced by multiple business and customer-related factors. The combination of exploratory analysis, statistical testing, and visualization provides a structured basis for understanding these patterns and supporting data-driven e-commerce decisions.

CHAPTER 14

FUTURE SCOPE

The current project provides a structured analysis of e-commerce revenue using data cleaning, transformation, feature engineering, exploratory data analysis, statistical analysis, visualization, and predictive modeling. The following areas can be considered for future development:

1. **Real-Time Revenue Analysis:** The analysis can be extended to real-time or frequently updated e-commerce data to monitor revenue, orders, customers, and sales performance continuously.
2. **Advanced Predictive Analysis:** Different machine learning models can be compared to improve revenue prediction and identify the model that provides the best performance for the available data.
3. **Customer Segmentation:** Customer-level information can be analyzed to identify different customer groups based on purchasing behavior, order value, frequency, and engagement.
4. **Improved Marketing Analysis:** Future analysis can examine the effectiveness of individual marketing campaigns, advertising channels, and promotional activities in greater detail.
5. **Advanced Time-Series Analysis:** Future work can use more detailed time-series techniques to study seasonal patterns, trends, and changes in revenue over longer periods.
6. **Interactive Business Dashboard:** The analysis can be converted into an interactive dashboard using tools such as Power BI or Tableau, allowing users to filter and monitor revenue by region, sales channel, time period, and other business factors.
7. **Additional Business Variables:** Future datasets can include product-level information, customer demographics, website traffic sources, payment methods, customer retention, returns, and competitor information to provide a more comprehensive analysis.
8. **Automated Reporting:** The analysis process can be automated so that updated data can be processed and new charts, statistics, and business insights can be generated with minimal manual effort.

Overall, the future scope of the project is to expand the analysis from historical revenue understanding toward continuous monitoring, deeper customer and marketing analysis, interactive reporting, and more accurate revenue prediction.

CHAPTER 15

DATASET AND OUTPUT SCREENSHOTS

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6000 entries, 0 to 5999
Data columns (total 23 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Record_ID                             6000 non-null   object
1   Date                                  6000 non-null   object
2   Marketing_Spend                       6000 non-null   float64
3   Website_Visits                       6000 non-null   int64
4   Unique_Visitors                      6000 non-null   int64
5   Conversion_Rate                      6000 non-null   float64
6   Average_Order_Value                  6000 non-null   float64
7   Orders                              6000 non-null   int64
8   Returning_Customer_Rate              6000 non-null   float64
9   Discount_Rate                       6000 non-null   int64
10  Ad_Clicks                           6000 non-null   int64
11  Email_Campaigns                     6000 non-null   int64
12  Mobile_Traffic_Percent              6000 non-null   float64
13  Customer_Rating                    6000 non-null   float64
14  Return_Rate                        6000 non-null   float64
15  Shipping_Delay_Days                 6000 non-null   int64
16  Product_Availability                6000 non-null   float64
17  Competitor_Price_Index              6000 non-null   float64
18  Seasonality_Index                  6000 non-null   float64
19  Economic_Index                     6000 non-null   float64
20  Sales_Channel                      6000 non-null   object
21  Region                             6000 non-null   object
22  Revenue                             6000 non-null   float64
dtypes: float64(12), int64(7), object(4)
memory usage: 1.1+ MB
```

```
numerical_summary = (
    df_clean[numeric_columns]
    .describe()
    .T
)

print(numerical_summary)
```

	count	mean	std	min	25%	\
Marketing_Spend	6000.0	185.105773	135.017081	12.42	95.7600	
Website_Visits	6000.0	900.623167	28.871279	800.00	881.0000	
Unique_Visitors	6000.0	640.015000	24.154186	559.00	624.0000	
Conversion_Rate	6000.0	3.820600	1.079726	0.80	3.1000	
Average_Order_Value	6000.0	46.725718	27.108174	5.12	27.7475	
Orders	6000.0	115.064667	10.300882	85.00	108.0000	
Returning_Customer_Rate	6000.0	38.095882	11.913369	8.00	29.8300	
Discount_Rate	6000.0	13.027500	8.253773	0.00	5.0000	
Ad_Clicks	6000.0	190.266333	13.490728	135.00	181.0000	
Email_Campaigns	6000.0	7.555500	4.594682	0.00	4.0000	
Mobile_Traffic_Percent	6000.0	56.952018	11.916908	20.00	48.7275	
Customer_Rating	6000.0	4.085718	0.525072	2.00	3.7300	
Return_Rate	6000.0	7.555812	3.483995	0.50	5.0900	
Shipping_Delay_Days	6000.0	1.512833	1.241273	0.00	1.0000	
Product_Availability	6000.0	90.667285	6.361308	67.11	86.3500	
Competitor_Price_Index	6000.0	100.075587	11.956356	70.00	92.0475	
Seasonality_Index	6000.0	1.051388	0.173792	0.75	0.9010	
Economic_Index	6000.0	100.017782	4.988119	79.33	96.6675	
Revenue	6000.0	6879.639398	4442.339107	566.16	3889.6600	

	50%	75%	max
Marketing_Spend	150.105	231.3325	1737.11
Website_Visits	900.000	920.0000	1000.00
Unique_Visitors	640.000	656.0000	728.00
Conversion_Rate	3.820	4.5500	7.73
Average_Order_Value	40.215	58.5900	231.69
Orders	115.000	122.0000	163.00
Returning_Customer_Rate	38.055	46.0725	75.00
Discount_Rate	10.000	20.0000	30.00
Ad_Clicks	190.000	199.0000	244.00
Email_Campaigns	8.000	11.0000	15.00
Mobile_Traffic_Percent	56.965	65.1925	90.00
Customer_Rating	4.100	4.4700	5.00
Return_Rate	7.480	9.8800	21.84
Shipping_Delay_Days	1.000	2.0000	7.00
Product_Availability	90.980	95.7600	100.00
Competitor_Price_Index	100.060	108.4650	135.00
Seasonality_Index	1.055	1.2030	1.35
Economic_Index	99.960	103.4400	120.67
Revenue	5763.245	8624.6725	50542.53

```
print(outlier_summary)
```

```
***
```

	Column	Q1	Q3	IQR	Lower_Bound	\
13	Shipping_Delay_Days	1.0000	2.0000	1.0000	-0.50000	
0	Marketing_Spend	95.7600	231.3325	135.5725	-107.59875	
18	Revenue	3889.6600	8624.6725	4735.0125	-3212.85875	
4	Average_Order_Value	27.7475	58.5900	30.8425	-18.51625	
2	Unique_Visitors	624.0000	656.0000	32.0000	576.00000	
8	Ad_Clicks	181.0000	199.0000	18.0000	154.00000	
3	Conversion_Rate	3.1000	4.5500	1.4500	0.92500	
1	Website_Visits	881.0000	920.0000	39.0000	822.50000	
17	Economic_Index	96.6675	103.4400	6.7725	86.50875	
5	Orders	108.0000	122.0000	14.0000	87.00000	
6	Returning_Customer_Rate	29.8300	46.0725	16.2425	5.46625	
14	Product_Availability	86.3500	95.7600	9.4100	72.23500	
10	Mobile_Traffic_Percent	48.7275	65.1925	16.4650	24.03000	
12	Return_Rate	5.0900	9.8800	4.7900	-2.09500	
11	Customer_Rating	3.7300	4.4700	0.7400	2.62000	
15	Competitor_Price_Index	92.0475	108.4650	16.4175	67.42125	
7	Discount_Rate	5.0000	20.0000	15.0000	-17.50000	
9	Email_Campaigns	4.0000	11.0000	7.0000	-6.50000	
16	Seasonality_Index	0.9010	1.2030	0.3020	0.44800	

	Upper_Bound	Outlier_Count	Outlier_Percentage
13	3.50000	422	7.033333
0	434.69125	326	5.433333
18	15727.19125	271	4.516667
4	104.85375	228	3.800000
2	704.00000	52	0.866667
8	226.00000	52	0.866667
3	6.72500	52	0.866667
1	978.50000	48	0.800000
17	113.59875	42	0.700000
5	143.00000	35	0.583333
6	70.43625	30	0.500000
14	109.87500	29	0.483333
10	89.89000	28	0.466667
12	17.06500	26	0.433333
11	5.58000	18	0.300000
15	133.09125	7	0.116667
7	42.50000	0	0.000000
9	21.50000	0	0.000000
16	1.65600	0	0.000000

BUSINESS FEATURE ANALYSIS

Revenue by Discount Category

```
discount_analysis = (  
    df_clean  
    .groupby(  
        'Discount_Category',  
        observed=True  
    )  
    .agg(  
        Count=('Revenue', 'size'),  
        Average_Revenue=('Revenue', 'mean'),  
        Average_Orders=('Orders', 'mean'),  
        Average_Order_Value=(  
            'Average_Order_Value',  
            'mean'  
        )  
    )  
)  
  
print(discount_analysis)
```

Discount_Category	Count	Average_Revenue	Average_Orders	Average_Order_Value
No Discount	669	7101.902212	115.112108	47.600284
Low Discount	2362	6843.517058	115.226503	46.067701
Medium Discount	2067	6847.344533	114.807934	46.800227
High Discount	902	6883.387328	115.194013	47.629424

Top 10 Numerical Relationships With Revenue

```
print(  
    "\nTop numerical relationships with Revenue:"  
)  
  
print(  
    revenue_correlations  
    .drop(  
        labels=target_derived_features,  
        errors='ignore'  
    )  
    .reindex(  
        revenue_correlations  
    )  
    .drop(  
        labels=target_derived_features,  
        errors='ignore'  
    )  
    .abs()  
    .sort_values(  
        ascending=False  
    )  
    .index  
)  
    .head(10)  
)
```

```
...  
Top numerical relationships with Revenue:  
Potential_Order_Value      0.922653  
Average_Order_Value        0.906231  
Discount_Adjusted_Order_Value 0.893239  
Seasonality_Index          0.247308  
Orders                     0.151443  
Visitor_Order_Rate         0.143262  
Orders_Per_Ad_Click        0.112180  
Availability_Risk          -0.101135  
Product_Availability        0.101135  
Estimated_Returning_Orders 0.070018  
Name: Revenue, dtype: float64
```

Revenue Distribution — Visualization

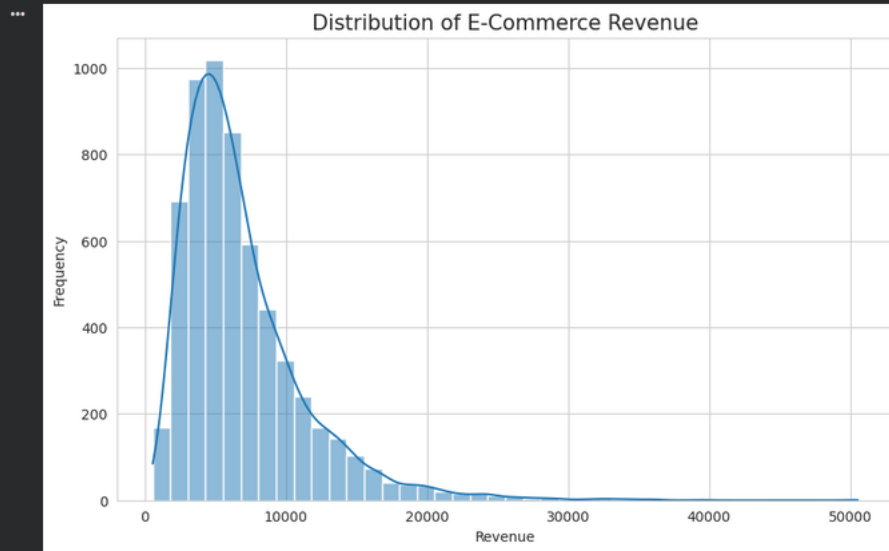
```
plt.figure(figsize=(10, 6))

sns.histplot(
    df['Revenue'],
    bins=40,
    kde=True
)

plt.title(
    'Distribution of E-Commerce Revenue',
    fontsize=15
)

plt.xlabel('Revenue')
plt.ylabel('Frequency')

plt.show()
```



Marketing Spend vs Revenue — Visualization

```
plt.figure(figsize=(9, 6))

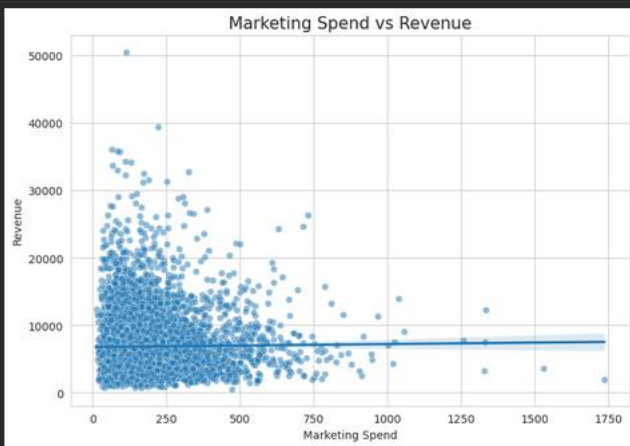
sns.scatterplot(
    data=df,
    x='Marketing_Spend',
    y='Revenue',
    alpha=0.5
)

sns.regplot(
    data=df,
    x='Marketing_Spend',
    y='Revenue',
    scatter=False
)

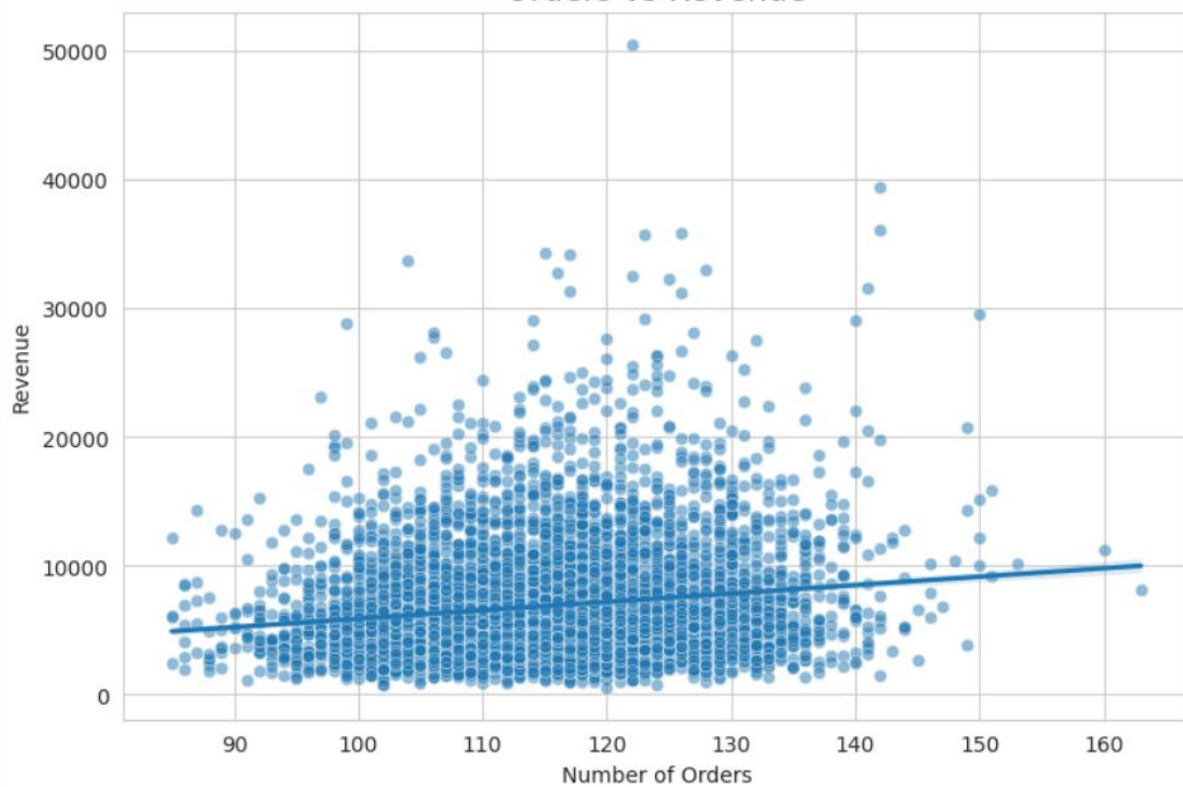
plt.title(
    'Marketing Spend vs Revenue',
    fontsize=15
)

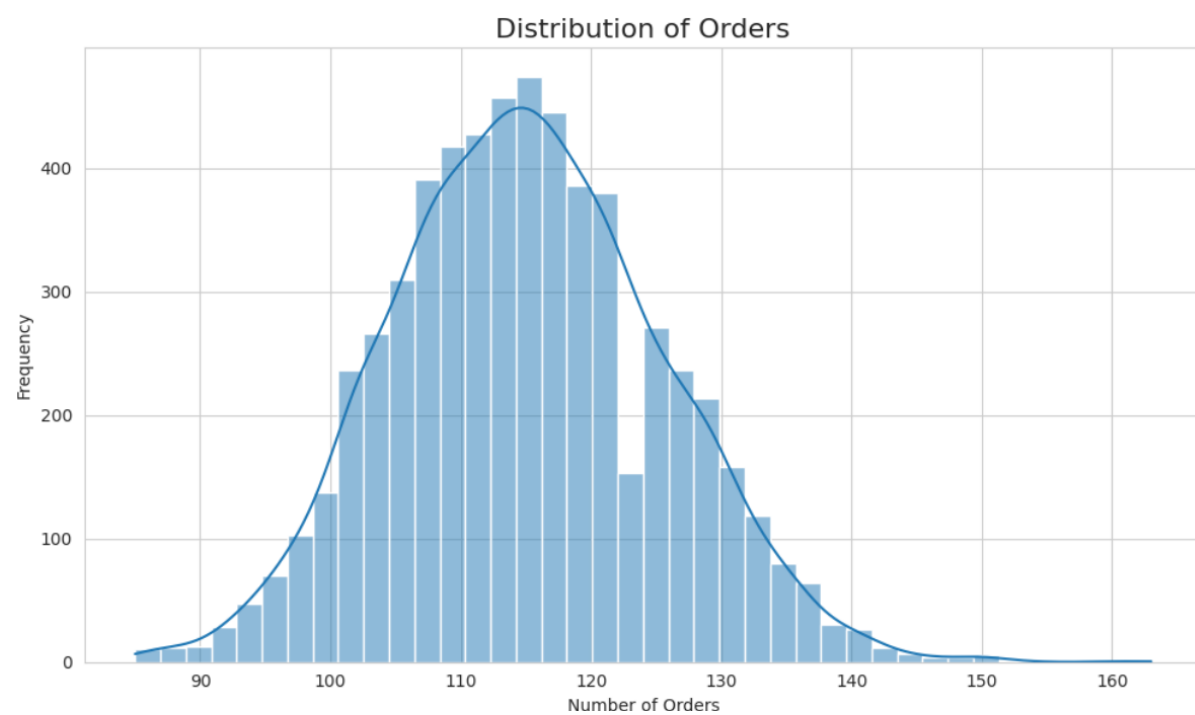
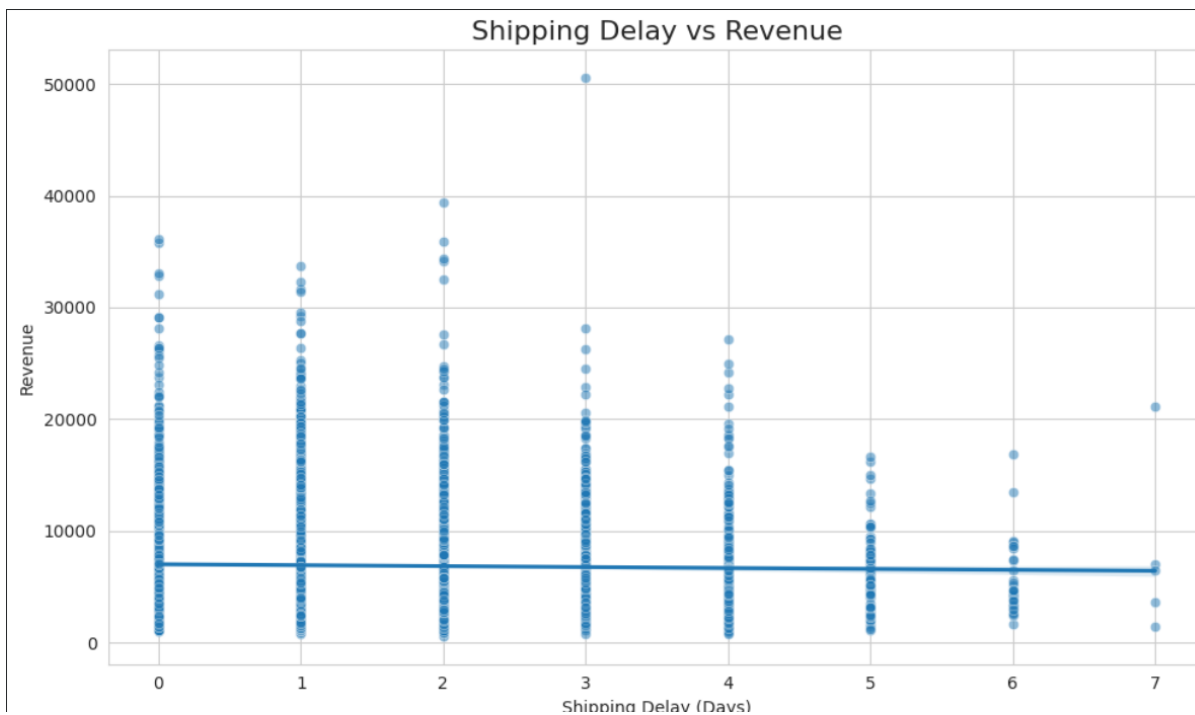
plt.xlabel('Marketing Spend')
plt.ylabel('Revenue')

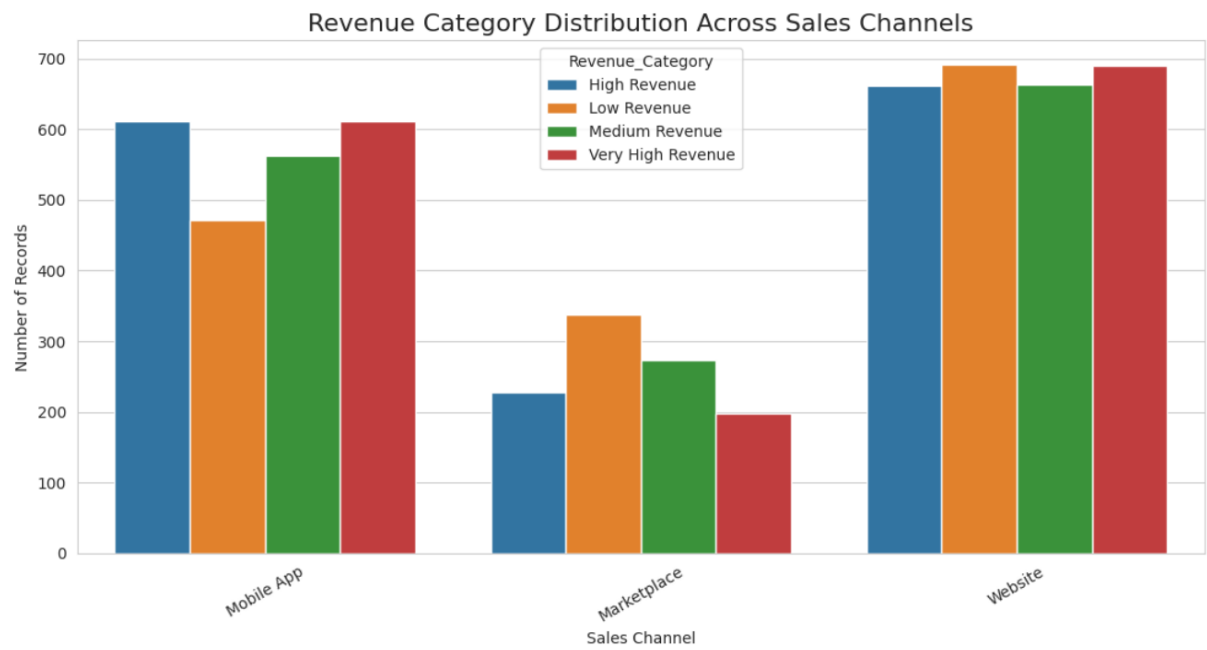
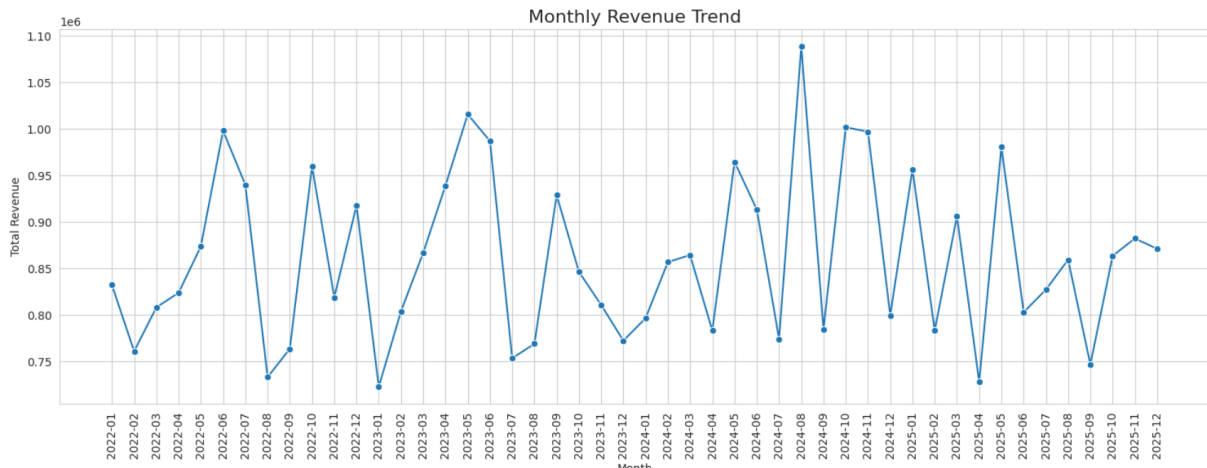
plt.show()
```



Orders vs Revenue







CHAPTER 16

CONCLUSION

The project “E-Commerce Revenue Analysis and Prediction” successfully analyzed historical e-commerce data to understand the factors and patterns associated with revenue performance. The analysis was carried out using Python through data cleaning, transformation, feature engineering, exploratory data analysis, statistical analysis, and data visualization.

The dataset containing 6,000 records from 2022 to 2025 was examined and prepared for analysis. Various features related to orders, customers, marketing, sales channels, regions, discounts, product availability, and time were analyzed to identify meaningful patterns. Feature engineering was also performed to create additional variables that supported deeper analysis.

The analysis showed that revenue was right-skewed, with an average revenue of approximately 6,879.64. The Mobile App recorded the highest average revenue among the sales channels, while the West region showed the highest average regional revenue. A strong positive relationship was observed between Average Order Value and Revenue, with a correlation of approximately 0.91. Marketing expenditure also showed a positive association with revenue.

Statistical analysis further supported several observations identified during exploratory analysis. Revenue differences across sales channels and regions were statistically significant, and the comparison between weekday and weekend revenue also showed a statistically significant difference at the 5% significance level.

The visualizations helped present revenue trends, distributions, relationships, and comparisons in a clear and understandable manner. These findings provide useful business insights that can support decisions related to sales channels, regional performance, customer purchasing behavior, marketing activities, and time-based promotions.

Overall, the project demonstrates how Python-based data analysis can be used to transform raw e-commerce data into meaningful information and business insights. The analysis provides a foundation for further predictive modeling, interactive dashboard development, customer analysis, and real-time revenue monitoring in future work.